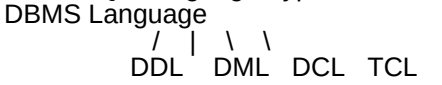
DBMS -- SQL Commands (DDL, DML, DCL, TCL)

Target: Name all 4 language types + their commands in 20s. Explain DROP vs TRUNCATE vs DELETE clearly.

The 4 SQL Language Types -- Overview



Type
Full Form
Purpose
Commands
Rollback?

DDL
Data Definition Language
Define/modify DB structure (schema)
CREATE, ALTER, DROP, TRUNCATE, RENAME
Auto-committed

DML
Data Manipulation Language
Manipulate data inside tables
SELECT, INSERT, UPDATE, DELETE
[OK] Can rollback

DCL
Data Control Language
Control access/permissions
GRANT, REVOKE
Auto-committed

TCL
Transaction Control Language
Manage transactions
COMMIT, ROLLBACK, SAVEPOINT
-- (controls others)

Key distinction to remember: DDL changes the structure. DML changes the data.

DDL -- Data Definition Language
DDL commands define or modify the schema (structure) of the database.
They are auto-committed -- you cannot rollback a DDL command.

CREATE
Used to create a new database or table.

```sql
-- Create a database
CREATE DATABASE Employee;
-- Show all databases
SHOW DATABASES;
-- Create a table
CREATE TABLE students (
  RoLL_No  INT(3),
  NAME     VARCHAR(20),
  SUBJECT  VARCHAR(20)
);
```

DROP
Used to permanently delete a database or table -- including all its data and structure.

Cannot be rolled back.
```sql
-- Drop a table
DROP TABLE table_name;
-- Drop an entire database
DROP DATABASE student_data;
```

DROP removes everything -- structure + data. No going back.

TRUNCATE
Used to remove all rows from a table but keep the table structure intact.
Faster than DELETE because it doesn't log individual row deletions.
```sql
TRUNCATE TABLE table_name;
-- Example: remove all rows from student_details
TRUNCATE TABLE student_details;
```

DROP vs TRUNCATE vs DELETE -- The Most Asked Comparison

Feature
DROP
TRUNCATE
DELETE

What it removes
Table structure + all data
All rows (keeps structure)
Selected rows (or all, with WHERE)

WHERE clause
No
No
[OK] Yes

Rollback possible
No (DDL)
No (DDL)
[OK] Yes (DML)

Speed
Fast
Very fast (no row-level log)
Slow (logs each row)

Resets AUTO_INCREMENT
[OK] Yes (table gone)
[OK] Yes
No

Triggers fired
No
No
[OK] Yes

Category
DDL
DDL
DML

Interview one-liner:

"DELETE removes specific rows and can be rolled back. TRUNCATE removes all rows fast but can't be rolled back -- it's DDL. DROP removes the entire table including structure -- also DDL, no rollback."

ALTER
Used to modify an existing table -- add, drop, or modify columns, or add constraints.

```sql
-- ADD a new column
ALTER TABLE table_name
ADD (column_name datatype);
-- Example: Add AGE and COURSE columns to student table
ALTER TABLE student
ADD (AGE NUMBER(3), COURSE VARCHAR(40));
-- DROP a column
ALTER TABLE table_name
DROP COLUMN column_name;
-- MODIFY an existing column's datatype
ALTER TABLE table_name
MODIFY column_name new_datatype;
```

DML -- Data Manipulation Language
DML commands read or modify the data inside tables.
They are not auto-committed -- you can rollback them using TCL commands.

SELECT
Fetch data from one or more tables.

```sql
-- Fetch specific column
SELECT column_name FROM table_name;
-- Fetch all columns
SELECT * FROM table_name;
-- Real examples
SELECT EMP_NAME FROM EMPLOYEES;
SELECT * FROM EMPLOYEES;
-- With condition
SELECT * FROM EMPLOYEES WHERE EMP_AGE &amp;gt; 25;
```

INSERT INTO
Add a new row into a table.

```sql
-- Method 1: Insert values for all columns (in order)
INSERT INTO table_name VALUES (value1, value2, value3);
-- Example
INSERT INTO student VALUES ('5', 'HARSH', 'WEST BENGAL', 19);
-- Method 2: Insert into specific columns only
INSERT INTO student (Roll_NO, Name, Age)
VALUES ('5', 'PRATIK', '19');
```

UPDATE
Modify existing data in a table.

```sql
-- Basic syntax
UPDATE table_name
SET column1 = value1, column2 = value2
WHERE condition;
-- Example 1: Set salary to 10000 for employees older than 25
UPDATE EMPLOYEES
SET EMP_SALARY = 10000
WHERE EMP_AGE &amp;gt; 25;
-- Example 2: Set specific employee salary
UPDATE EMPLOYEES
SET EMP_SALARY = 120000
WHERE EMP_NAME = 'APOORV';
```

Always use WHERE with UPDATE -- without it, every row gets updated.

DELETE
Remove specific rows from a table.

```sql
-- Delete with condition
DELETE FROM table_name WHERE condition;
-- Example: delete all employees in HR dept
DELETE FROM EMPLOYEES WHERE DEPT = 'HR';
-- Delete ALL rows (can be rolled back -- unlike TRUNCATE)
DELETE FROM table_name;
```

DCL -- Data Control Language
Controls who can do what in the database.
GRANT
Give a user access/privileges to a database object.
sql
GRANT SELECT, INSERT ON students TO user_name;
GRANT ALL PRIVILEGES ON database_name TO user_name;
REVOKE
Take back privileges previously given via GRANT.
sql
REVOKE SELECT ON students FROM user_name;

TCL -- Transaction Control Language
Controls the outcome of DML transactions (groups of SQL operations).
COMMIT
Permanently save all changes made in the current transaction.
sql
COMMIT;
-- After COMMIT, changes cannot be rolled back
ROLLBACK
Undo all changes made since the last COMMIT or SAVEPOINT.
sql
ROLLBACK;
-- Reverts all uncommitted DML changes
SAVEPOINT
Set a checkpoint within a transaction. You can rollback to a specific savepoint instead of rolling back everything.
```sql
SAVEPOINT savepoint_name;
-- Example
INSERT INTO students VALUES (1, 'Avanish');
SAVEPOINT sp1;
INSERT INTO students VALUES (2, 'Rahul');
ROLLBACK TO sp1;   -- undoes Rahul's insert; Avanish's insert is safe
COMMIT;            -- saves Avanish permanently
```

TCL Transaction Flow -- Visual
```
BEGIN (transaction starts automatically with first DML)
  INSERT / UPDATE / DELETE  (DML operations)

  SAVEPOINT sp1             (optional checkpoint)

  more DML...

  ROLLBACK TO sp1           (undo back to checkpoint)
    OR
  COMMIT                  (save everything permanently)
    OR
  ROLLBACK                  (undo everything since last COMMIT)

```

Quick Cheat Sheet
```
DDL   CREATE, ALTER, DROP, TRUNCATE, RENAME    changes structure   auto-committed
DML   SELECT, INSERT, UPDATE, DELETE           changes data       can rollback
DCL   GRANT, REVOKE                          controls access    auto-committed
TCL   COMMIT, ROLLBACK, SAVEPOINT              controls txn       wraps DML
DROP    = delete table + structure (no rollback)
TRUNCATE = delete all rows, keep structure (no rollback, very fast)
DELETE  = delete specific rows (can rollback, fires triggers)
```

Your 40-Second Script -- SQL Languages

"SQL is divided into 4 categories. DDL -- Data Definition Language -- defines the structure:
CREATE, ALTER, DROP, TRUNCATE. It's auto-committed so you can't rollback DDL.
DML -- Data Manipulation Language -- handles data: SELECT, INSERT, UPDATE, DELETE.
DML can be rolled back. DCL controls permissions -- GRANT and REVOKE.
TCL manages transactions -- COMMIT saves permanently, ROLLBACK undoes changes,
SAVEPOINT sets a checkpoint mid-transaction."

Follow-Up Questions (Expect These)
Q: Can you rollback a DROP statement?

No. DROP is DDL and is auto-committed. Once dropped, the table is gone unless you have a backup.

Q: What happens if you UPDATE without a WHERE clause?

Every row in the table gets updated. Always double-check your WHERE clause before running UPDATE or DELETE.

Q: What is the difference between GRANT and REVOKE?

GRANT gives privileges to a user. REVOKE takes them back. Both are DCL commands and auto-committed.

Q: Can you ROLLBACK after a COMMIT?

No. COMMIT makes the changes permanent and durable. You cannot undo it with ROLLBACK -- only a new DML operation can reverse it.